A Fredholm Ownership Firewall for the Weighted Hénon Reflection Euler Family

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Abstract

The weighted reflection Euler family of the full Hénon horseshoe has an explicit punctured scalar continuation. We ask whether a Fredholm determinant representation gives that continuation meaningful operator ownership. Its logarithmic channels can indeed be placed on the diagonal of a locally trace-class holomorphic family $A(z,q)$, and $\det_{\mathrm{F}}(\exp A) = \exp(\operatorname{Tr} A)$ recovers the continuation exactly. This realization is nevertheless tautological: every nonvanishing holomorphic scalar F satisfies $F = \det_{\mathrm{F}}(I+(F-1)P)$ for a fixed rank-one projection P. In the opposite direction, each primitive source word owns a finite weighted cyclic block B_ω with $\det(I - zB_\omega) = 1 - z^n q^{S_n\chi(\omega)}$. This is the P70 Euler denominator, whose reciprocal is the corresponding Euler factor. Its singular values are the physical edge weights in $\{1, q\}$. Primitive singleton reflection words at every odd length therefore put a uniform positive singular-value floor in the undamped full orbit-block direct sum. That sum is bounded but noncompact, belongs to no finite Schatten ideal, and has no ordinary trace-class Fredholm determinant. The comparison furnishes a claim firewall: analytic determinant realization is proved but tautological, finite source blocks are exact, and a genuine source-derived transfer owner remains open.

1 The operator question after scalar continuation

The Hénon horseshoe admits a full-shift model in the hyperbolic regime relevant here [2, 1]. Previous steps of the present programme associated to its marked reflection packets the weighted channel expansion

$$\mathcal{L}(z,q) = \sum_{m\geq 1} h_m(z,q), \qquad h_m(z,q) = c_m \frac{2(qz)^m}{1-(1+q^{2m})z^{2m}}, \tag{1}$$

where

$$c_m = \frac{1}{m} \prod_{\substack{p|m \\ p \text{ odd}}} (1-p), \qquad 0 < |c_m| \leq 1. \tag{2}$$

For fixed $q > 0$, put

$$L(q) = \min(1, q^{-1}), \tag{3}$$

$$\rho_m(q) = (1+q^{2m})^{-1/(2m)}, \tag{4}$$

$$\Sigma_q = \{\rho_m(q)e^{\pi i k/m} : m \geq 1,\ 0 \leq k < 2m\}, \tag{5}$$

$$\Omega_q = \{z : |z| < L(q)\} \setminus \Sigma_q. \tag{6}$$

The preceding scalar theory proves that (1) converges normally on compact subsets of Ω_q and defines the nonvanishing continuation

$$\mathcal{Z}_{\mathrm{ch}}(z,q) = \exp \mathcal{L}(z,q). \tag{7}$$

It also classifies the limiting circle; we do not revisit or cross that boundary. Instead we distinguish two questions:

1. Can the known scalar (7) be represented by a legitimate trace-class Fredholm determinant?
2. Does an operator built independently from the source dynamics own the same determinant?

The first answer is yes. The second is not implied by the first.

2 A locally trace-class channel realization

Let $(e_m)_{m\geq 1}$ be the standard basis of $\ell^2(\mathbb{N})$ and let P_m project onto $\mathbb{C}e_m$. Define

$$A(z,q) = \sum_{m\geq 1} h_m(z,q)P_m = \mathrm{diag}(h_1(z,q), h_2(z,q), \ldots). \tag{8}$$

Theorem 2.1 (Channel-diagonal determinant)**.** *For every fixed $q > 0$, $A : \Omega_q \to \mathcal{S}_1$ is holomorphic in trace norm. The family*

$$K_{\mathrm{ch}}(z,q) = \exp A(z,q) - I$$

is likewise trace-class holomorphic, and

$$\det_{\mathrm{F}}(I + K_{\mathrm{ch}}(z,q)) = \exp(\mathrm{Tr}\, A(z,q)) = \mathcal{Z}_{\mathrm{ch}}(z,q). \tag{9}$$

Proof. Let $C \Subset \Omega_q$. Choose $r < L(q)$ such that $|z| \leq r$ on C. Then $r < 1$ and $qr < 1$. All finitely many early denominators in (1) are uniformly separated from zero on C. For all sufficiently large m and all $z \in C$,

$$\left|1 - (1+q^{2m})z^{2m}\right| \geq 1 - r^{2m} - (qr)^{2m} \geq \frac{1}{2}.$$

Using (2),

$$\sup_{z\in C} |h_m(z,q)| \leq 4(qr)^m \tag{10}$$

for the same tail. Thus $\sum_m h_m P_m$ converges locally uniformly in $\mathcal{S}_1$, since the trace norm of a diagonal operator is the sum of the absolute diagonal entries. The Banach-valued Weierstrass theorem gives trace-norm holomorphy, and $\mathrm{Tr}\, A = \sum_m h_m = \mathcal{L}$.

The exponential remains in $I + \mathcal{S}_1$. More explicitly, for the diagonal tail,

$$|e^{h_m} - 1| \leq e^{|h_m|}|h_m|,$$

so (10) again gives locally uniform trace-norm convergence. The standard trace-ideal identity $\det_{\mathrm{F}}(e^A) = e^{\mathrm{Tr}\, A}$ [3] now yields (9). □

This theorem is an exact analytic statement. It is not yet a test of dynamical provenance, because the entries of A were read from the already completed scalar logarithm.

3 The universal rank-one firewall

Lemma 3.1 (Universal scalar realization)**.** *Let $U \subset \mathbb{C}$ be a domain, let F be a nonvanishing holomorphic function on U, and let P be a fixed rank-one orthogonal projection on an arbitrary Hilbert space. Then*

$$K_F(z) = (F(z) - 1)P$$

is trace-class holomorphic and

$$\det_{\mathrm{F}}(I + K_F(z)) = F(z), \qquad z \in U. \tag{11}$$

Proof. Scalar multiplication makes K_F holomorphic in trace norm. It has one possibly nonzero eigenvalue, $F(z) - 1$. The finite-rank definition of the Fredholm determinant therefore gives $\det_{\mathrm{F}}(I + K_F) = 1 + (F - 1) = F$. □

Lemma 3.1 is deliberately elementary. It shows that the bare existence of a parameter-dependent trace-class family with determinant F is universal and therefore cannot, by itself, identify a transfer mechanism, periodic-point trace, or dynamically selected state space. We call such a construction a *scalar-built realization*. A genuine source owner would have to be specified from the dynamics independently of the already known F and then derive F as a consequence. In this sense Theorem 2.1 is `PROVED_TAUTOLOGICAL`: correct and useful for analysis, but insufficient as an ownership theorem.

4 Finite source-native cyclic blocks

The literal source construction has more provenance. Let ω be a primitive marked reflection word of odd length n. Write

$$\chi_j = \chi(\sigma^j \omega) \in \{0, 1\}, \qquad S(\omega) = \sum_{j=0}^{n-1} \chi_j.$$

On $\mathbb{C}^n$, with indices read modulo n, set

$$B_\omega e_j = q^{\chi_j} e_{j+1}. \tag{12}$$

These are the physical edge weights; no period damping has been inserted.

Proposition 4.1 (Exact orbit block)**.** *For every $q > 0$ and every such ω,*

$$B_\omega^n = q^{S(\omega)} I, \tag{13}$$

$$\det(I - zB_\omega) = 1 - z^n q^{S(\omega)}, \tag{14}$$

and the singular values of B_ω are precisely the multiset $\{q^{\chi_0}, \ldots, q^{\chi_{n-1}}\} \subset \{1, q\}$. Consequently

$$\min(1, q)\|x\| \leq \|B_\omega x\| \leq \max(1, q)\|x\|. \tag{15}$$

Proof. After one full cycle, every e_j has acquired every edge weight once, which proves (13). Moreover $e_0, B_\omega e_0, \ldots, B_\omega^{n-1} e_0$ are nonzero multiples of the standard basis, so e_0 is cyclic. Hence the minimal polynomial has degree n; by (13) it is $\lambda^n - q^{S(\omega)}$, which is also the characteristic polynomial. Substitution gives (14). Finally,

$$B_\omega^* B_\omega e_j = q^{2\chi_j} e_j.$$

Taking positive square roots proves the singular-value statement and (15). □

Thus every finite Euler denominator has an exact source-native owner, and the corresponding P70 Euler factor is the reciprocal determinant $\det(I - zB_\omega)^{-1}$. The issue is whether all of these blocks are summable in a determinant class.

5 The full source sum is not compact

For every odd $n \geq 3$, let $\omega_n = (1, 0, \ldots, 0)$ with the singleton at the chosen reflection center. Reflection fixes ω_n. It is primitive: repetition of a shorter word would create more than one symbol equal to 1. Exactly the two sites adjacent to the singleton have unequal distance-two neighbours, so

$$S(\omega_n) = n - 2. \tag{16}$$

Let $\mathfrak{R}$ be the countable collection of physical primitive marked reflection words and form the Hilbert direct sum

$$\mathcal{H} = \bigoplus_{\omega \in \mathfrak{R}} \mathbb{C}^{|\omega|}, \qquad B_q = \bigoplus_{\omega \in \mathfrak{R}} B_\omega. \tag{17}$$

Theorem 5.1 (Source-native obstruction)**.** *For every $q > 0$, B_q is bounded and noncompact. It belongs to no Schatten class $\mathcal{S}_p$ with $0 < p < \infty$. In particular, for $z \neq 0$ the expression $\det_{\mathrm{F}}(I - zB_q)$ is not an ordinary trace-class Fredholm determinant.*

Proof. The uniform bounds (15) pass to the Hilbert direct sum:

$$\min(1, q)\|x\| \leq \|B_q x\| \leq \max(1, q)\|x\|.$$

Choose one unit basis vector u_n from each singleton block. The vectors u_n are orthonormal, their images lie in mutually orthogonal blocks, and $\|B_q u_n\| \geq \min(1, q) > 0$. Hence $(B_q u_n)$ has no convergent subsequence, so B_q is not compact. Every finite Schatten class is contained in the compact operators. Thus $B_q \notin \mathcal{S}_p$ for every finite p; in particular $zB_q \notin \mathcal{S}_1$ when $z \neq 0$, which excludes the ordinary trace-class determinant. $\square$

The theorem concerns the literal undamped source blocks. It does not refute every conceivable transfer operator. It says that compactness or nuclearity must come from additional source-derived structure, rather than from taking the naive full direct sum.

6 A graded ledger is not an operator trace

There is a useful but dangerous remnant of trace combinatorics. From the cyclic shift action,

$$\sum_{j=0}^{n-1} \langle e_j, B_\omega^r e_j \rangle = \begin{cases} 0, & n \nmid r, \\ nq^{(r/n)S(\omega)}, & n \mid r. \end{cases} \tag{18}$$

For fixed r, only the finitely many lengths $n \mid r$ contribute, and there are finitely many binary source words at each such length. Therefore (18) reproduces a locally finite coefficient ledger after summing over blocks.

This does not contradict Theorem 5.1. The left side of (18) is a finite-block diagonal sum; its aggregate is tied to the canonical block basis. Since B_q^r is still bounded below and noncompact, it is not trace class. The aggregate cannot be promoted to a basis-independent Hilbert-space trace or used in the Fredholm trace expansion. We retain it only as a formal graded identity.

7 Executable certificate and claim boundary

The accompanying standard-library certificate performs exact rational Gaussian elimination on finite cyclic blocks, verifies representative weighted full turns, compares channel trace exponentials with products of diagonal exponentials, and tests unrelated rank-one realizations. A second program reconstructs the formulas without importing the primary module. Normal and optimized unit-test runs enforce the same schema, and mutation tests reject promotions of the operator and arithmetic claims. These computations audit finite identities; the local trace-norm and noncompactness arguments remain the proofs.

Claim	Status
Punctured analytic determinant	`PROVED_TAUTOLOGICAL`
Universal rank-one firewall	`PROVED`
Finite source cyclic blocks	`PROVED`
Full source direct sum trace class	`REFUTED`
Genuine source-derived transfer owner	`OPEN`
Arithmetic trace or Route B	`NO / NOT AUTHORIZED`

No rational-prime semantics, von Mangoldt trace, self-adjoint Hilbert–Pólya operator, or arithmetic advance follows. The constructive next question is precise: find a dynamically forced state space and a source-derived compactness mechanism whose genuine operator traces produce the weighted channels without inserting the completed scalar function.

References

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